

Where Do Major U.S. Airlines Lose On-Time Performance to Causes They Control?

A controllable-disruption peer benchmark of American, Delta, United, JetBlue and Southwest — July 2025 to June 2026

Final report · PACE framework (Plan, Analyze, Construct, Execute)

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Data: U.S. DOT, Bureau of Transportation Statistics — Reporting Carrier On-Time Performance (4,471,010 flights of the five client airlines)

Repository: <https://github.com/donyvaldez/airline-delay-analysis>

Interactive dashboard: [Tableau Public](#)

1. Executive summary

Question. Where do the five largest U.S. airlines lose on-time performance to causes within their own control, and what can their operations teams do about it?

Approach. Twelve months of official flight-level data (July 2025 – June 2026) were validated and analyzed. "Controllable" disruption was defined as carrier-caused delay minutes and carrier-caused (code A) cancellations, following DOT Technical Reporting Directive #40. Each airline was compared with the other client airlines **departing from the same airport**, so that weather, runways and air traffic control were shared conditions.

Key findings

1. **American at Dallas/Fort Worth is the largest controllable problem.** 25.2% of AA's DFW flights arrived late with carrier-caused minutes, vs. 8.8% for peers at the same airport — about **26,100 excess controllable late flights per year**, and worse than peers in all 12 months.
2. **AA's problem starts before any delay can propagate.** 18.0% of AA's first flights of the day at DFW departed 15+ minutes late, vs. 5.1%–9.7% for the other airlines at their main airports.
3. **Scheduled ground time explains recovery.** When an aircraft arrives late, airlines that schedule more time between flights recover more delay on the ground. The ranking holds for all five airlines. At Denver, United schedules 74 minutes and recovers 20.8; Southwest schedules 50 and recovers none.
4. **Southwest at Denver is an early warning.** Its controllable gap vs. peers widened in April–June 2026, its three worst months of the year.

Recommendations

#	Airline	Recommendation
R1	AA at DFW	Diagnose morning station readiness with internal data and bring first-flight punctuality toward the peer level
R2	AA at DFW	Reduce late-arriving aircraft into the hub and review afternoon and evening ground times
R3	WN at DEN	Pilot longer ground times on part of the 12:00–17:59 rotations before extending them
R4	B6 at BOS	Review the ground turnaround process, with a focus on winter readiness
R5	DL and UA	Maintain current practices; UA at DEN is the internal benchmark; DL should review its carrier-cancellation rate
P1	AA and DL	Open question: opposite "operate late" vs. "cancel" strategies? Requires internal data

The analysis is reproducible and the dashboard can be refreshed monthly as BTS publishes new data, turning a one-time study into a monitoring tool.

2. Business context

Item	Definition
Client	Operations executive teams of American (AA), Delta (DL), United (UA), JetBlue (B6) and Southwest (WN), in a peer benchmark report
Decision supported	Where (airport, hour of day, month) each airline's controllable cancellations and delays exceed those of its peers, to focus staffing, maintenance and turnaround improvements
Success criteria	Rank the airlines by on-time rate (arrival < 15 min late, DOT standard), cancellation rate and controllable share; identify where each is worse than peers; separate controllable from non-controllable causes
Scope	Scheduled, nonstop, domestic passenger flights, July 2025 – June 2026, operated under each airline's own reporting code
Out of scope	International flights, cargo, charters, fares, passengers per flight and regional affiliates (Envoy, PSA, SkyWest, Republic)

3. Data and methodology

3.1 Data source

BTS TranStats — Reporting Carrier On-Time Performance: one row per flight, published monthly by the U.S. Department of Transportation. Reporting is mandatory for carriers above 0.5% of domestic scheduled passenger revenue (14 carriers in 2026, including all five clients). The data is public domain and contains no personal information.

3.2 Validation

No record was deleted or altered. Eight tests were run (Appendix A). The most important: for 177,012 of 177,012 late flights in July 2025 (100%), the five delay-cause fields added up exactly to the arrival delay, as DOT rules require. Row counts reconciled across every phase (631,428 flights in July; 4,471,010 client flights over 12 months).

3.3 Key definitions

Term	Definition
Late	Arrival 15 or more minutes after schedule (DOT standard)
Controllable	Carrier-caused delay minutes (CarrierDelay) and code-A cancellations. A project category grounded in Directive #40, not an official DOT field
Controllable late rate	Operated flights late with at least one carrier-caused minute ÷ operated flights
Peers	The other client airlines departing the same airport; pooled average (total events ÷ total flights)
Main airport	Each airline's airport with the most departures: AA-DFW, DL-ATL, UA-DEN, WN-DEN, B6-BOS
Excess late flights	Operated flights × gap vs. peers ÷ 100: late flights above the peer level (an estimate)
Recovery	Delay the aircraft arrived with minus delay its next flight departed with (same aircraft, same day)

Full definitions of all columns, codes and metrics are in ``docs/data_dictionary.md``.

3.4 Tools

Python (pandas) in Jupyter for ingestion, validation and analysis; Tableau Public for the dashboard; GitHub for version control and documentation. BigQuery was originally planned but not needed: the client-airline dataset (about 1 GB in memory) was analyzed entirely in pandas.

4. Step-by-step process

Phase 1 — Plan

- Defined the client, the decision, the success criteria and the scope before touching the data.
- Chose the official BTS source over a packaged 2015 Kaggle dataset, which was outdated for a 2025–2026 question.
- Built ``01_download_bts.py``: downloads each month, verifies the ZIP, keeps 31 columns, saves Parquet and logs quality checks. All 12 months passed.

Phase 2 — Analyze

- Explored one month (July 2025) column by column; learned that empty delay-cause fields on on-time flights are missing by design, not errors.
- Ran control test 1: delay causes reconcile to arrival delay for 100% of late flights.
- Compared airlines by late rate (not counts), then by route, then by airport. Found that the worst routes concentrate at AA's DFW hub.
- Introduced the same-airport peer benchmark and confirmed it over 12 months: AA at DFW and WN at DEN were worse than peers every month.

Phase 3 — Construct

- Rebuilt the benchmark with controllable causes only, over 12 months.
- Replaced an obvious hypothesis ("late arrivals cause late departures") with decision-useful ones: H2 recovery, H3 schedule buffer, H4 aircraft utilization. Tested them on 3,044,604 consecutive flight pairs.
- Analyzed time of day and tested the assumption that pre-09:00 departures are first flights. The assumption was **rejected** (only 43%–89% are), so the reported metric was changed to the first-flight late rate.
- Exported five result tables and published a five-chart dashboard on Tableau Public.

Phase 4 — Execute

- Translated findings into recommendations with evidence, a measure of success and a cost or risk for each; wrote this report and an executive presentation.

5. Findings

5.1 Who has the controllable problem

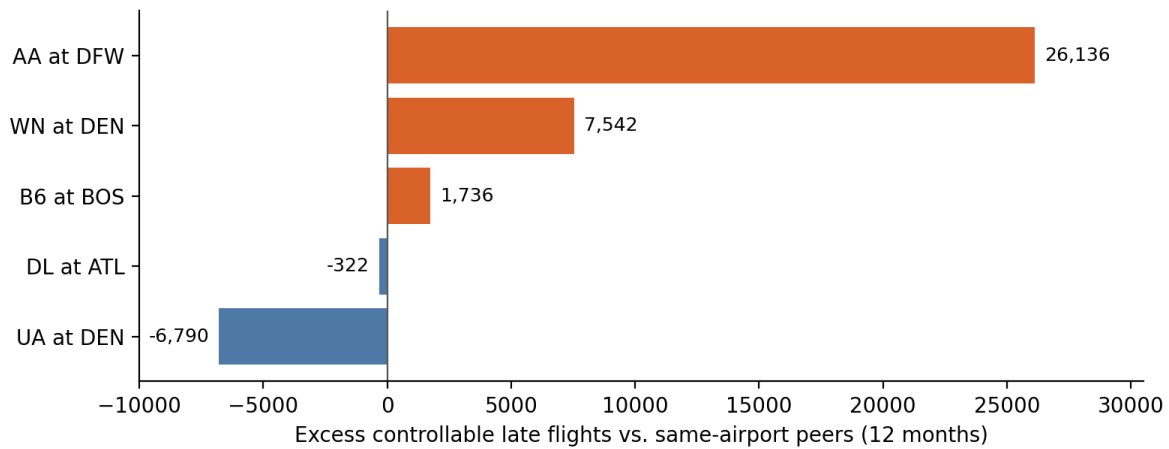


Figure 1. Excess controllable late flights vs. same-airport peers, each airline at its main airport (12 months)

Airline at main airport	Controllable late rate	Peers	Gap (pts)	Carrier min/flight (peers)	Carrier cancel % (peers)
AA at DFW	25.2%	8.8%	+16.4	8.1 (5.8)	0.25 (0.83)
WN at DEN	20.5%	12.0%	+8.6	5.3 (4.3)	0.14 (0.20)
B6 at BOS	17.1%	12.3%	+4.8	5.1 (5.3)	0.38 (0.48)
DL at ATL	13.2%	13.3%	-0.1	5.7 (4.7)	0.98 (0.24)
UA at DEN	12.2%	18.6%	-6.4	3.7 (5.7)	0.14 (0.22)

American's excess at DFW is 3.5 times the next case. Southwest is above its peers at every airport where it appears (DEN, BOS, ATL). Caution: at a hub the peer pool is dominated by the host airline, so the most reliable rows are each airline at its own main airport (full table in Appendix C).

5.2 Is it permanent?

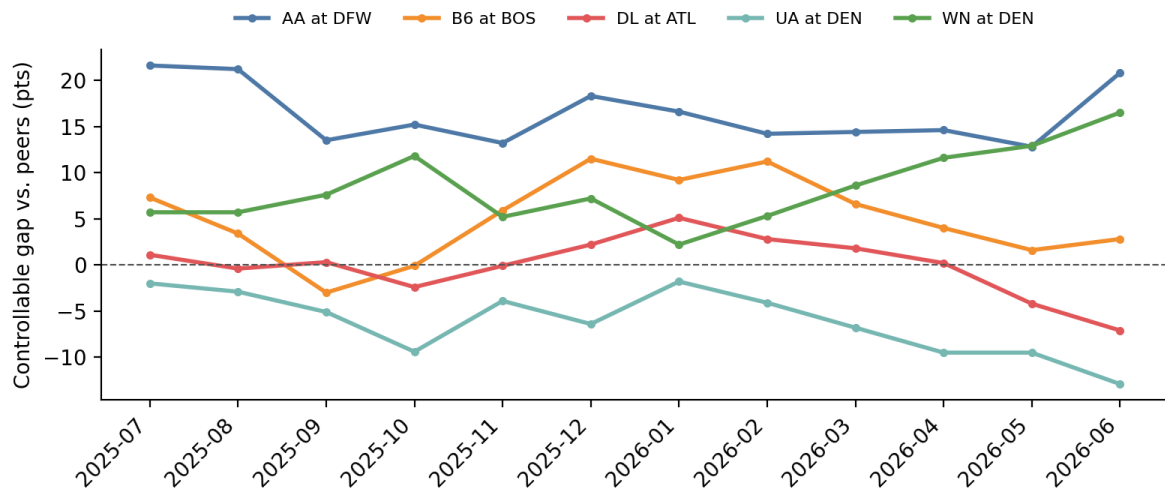


Figure 2. Monthly controllable gap vs. peers (percentage points)

Airline	Months worse than peers	Average monthly gap	Pattern
AA at DFW	12 / 12	+16.4	Structural — never below +12.8
WN at DEN	12 / 12	+8.4	Worsening in Apr–Jun 2026
B6 at BOS	10 / 12	+5.0	Seasonal (winter)
DL at ATL	7 / 12	-0.1	Neutral
UA at DEN	0 / 12	-6.2	Better than peers every month

5.3 Why some airlines recover and others do not

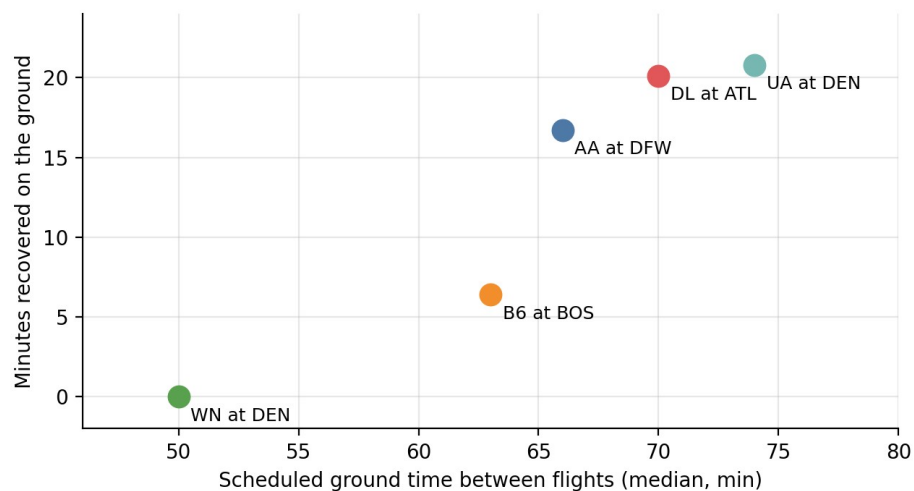


Figure 3. Scheduled ground time vs. minutes recovered after a late inbound aircraft (main airports, 12 months)

Airline	Ground time (median)	Inbound late	Recovered	Departs on time after late inbound	Legs per aircraft-day
WN at DEN	50 min	23.9%	0.0 min	13.3%	5.0

Airline	Ground time (median)	Inbound late	Recovered	Departs on time after late inbound	Legs per aircraft-day
B6 at BOS	63 min	25.1%	6.4 min	21.7%	2.9
AA at DFW	66 min	21.9%	16.7 min	29.8%	3.4
DL at ATL	70 min	12.4%	20.1 min	36.7%	3.7
UA at DEN	74 min	16.3%	20.8 min	43.3%	3.0

- **H2 supported:** recovery ranges from none (WN) to 20.8 minutes (UA).
- **H3 supported:** recovery ranks in the same order as scheduled ground time for all five airlines — a strong, consistent relationship, not proof of causation.
- **H4 not supported:** DL flies more legs per aircraft-day than AA and still recovers more.
- **AA at DFW** receives late aircraft almost twice as often as DL at ATL and recovers less with a similar buffer — pointing to ground execution as well as inbound delays.

5.4 When to act: time of day and the first flight

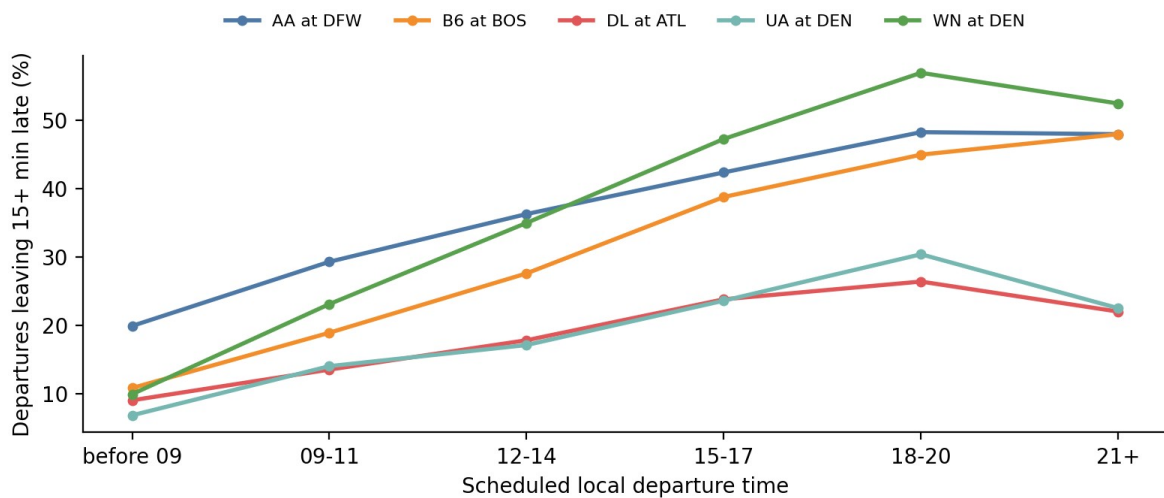


Figure 4. Departures leaving 15+ minutes late by scheduled local departure time (main airports)

- Delay accumulates through the day for every airline. WN has the steepest rise (9.9% → 57.0%) and schedules its shortest ground times (45 min) exactly between 12:00 and 17:59, where its recovery turns negative.
- UA at DEN widens ground time to 82 minutes at midday and 97 at night and recovers 39–40 minutes in those bands: the best-practice benchmark.
- AA at DFW keeps a flat schedule (64–66 min) all day and does not adapt to afternoon accumulation.

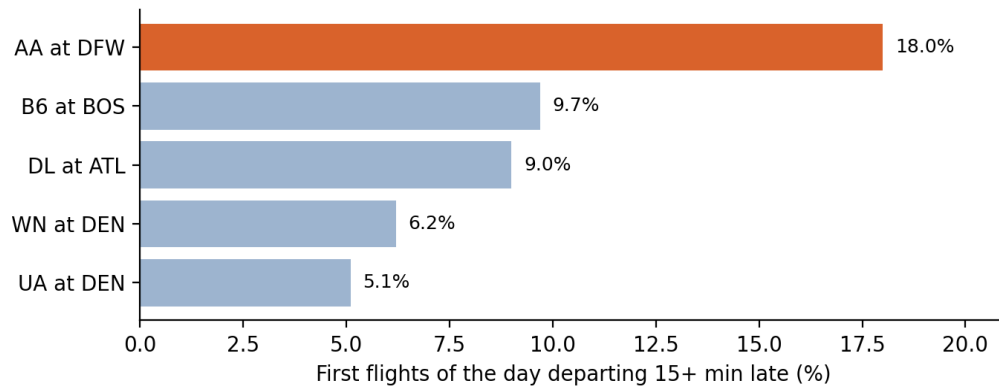


Figure 5. First flight of the day departing 15+ minutes late — no inbound aircraft, so no propagation

Because a first flight has no inbound aircraft, its delays reflect station readiness — the purest controllable signal. AA's rate (18.0%) is about 2 times DL's and 3.5 times UA's.

5.5 An open pattern: cancel or fly late?

AA has the lowest carrier-cancellation rate at DFW (0.25% vs. 0.83% for peers) and the largest controllable delay. DL cancels for carrier reasons at about four times its peers' rate at ATL (0.98% vs. 0.24%) with neutral controllable delay, and above its peers at all four airports. This is consistent with opposite operational strategies, but the data cannot show internal decisions; it is reported as an open question.

6. Recommendations

#	Action	Evidence	How to measure	Cost or risk
R1 • AA at DFW	Diagnose morning station readiness (crew, overnight maintenance, ramp) with internal data; bring first-flight punctuality toward the peer level	18.0% of first flights depart late vs. 5.1%–9.7% for others; no inbound aircraft, so no propagation	First-flight late rate (target: toward DL's 9%)	Low: internal diagnosis before investment
R2 • AA at DFW	Reduce late-arriving aircraft into the hub and review afternoon and evening ground times	Receives late aircraft 21.9% of the time (DL: 12.4%); flat 64–66 min schedule while punctuality falls from 20% to 48%	Minutes recovered on the ground; monthly controllable gap	Longer ground times mean fewer flights per aircraft per day
R3 • WN at DEN	Pilot longer ground times on part of the 12:00–17:59 rotations; measure before extending	45 min scheduled in those bands with negative recovery; UA at the same airport schedules 74–82 and recovers 20–40; gap widening	Recovery in the 12:00–17:59 bands; monthly gap trend	Business model relies on quick turns; a pilot limits the cost
R4 • B6 at BOS	Review the ground turnaround process, focused on winter readiness	Recovers only 1.9–7.8 min with buffers similar to DL's; seasonal gap (Dec–Mar)	Minutes recovered; winter gap	Low: process review
R5 • DL and UA	Maintain practices; use UA at DEN as the internal benchmark; DL to review its carrier-cancellation rate	UA better than peers 12 of 12 months; DL cancels about 4x its peers at ATL	Monthly gap; code-A cancellation rate	—
P1 • AA and DL	Open question: "operate late" vs. "cancel" strategies	AA cancels least and delays most; DL shows the opposite	Requires internal airline data	Presented as a question, not a conclusion

Monitoring. The ingestion script and notebooks are reproducible, so the benchmark and dashboard can be refreshed each month to track whether the actions work.

7. Limitations

1. Staffing and maintenance cannot be separated: both are grouped inside carrier-caused delay.
2. Delay causes are self-reported by airlines under DOT rules.
3. Regional affiliates are excluded; results reflect mainline operations only.
4. Each airline is compared at its single busiest airport; secondary hubs are not analyzed.
5. At a hub, the peer pool is dominated by the host airline.
6. Aircraft rotations are followed within the same day only; overnight propagation is not measured.
7. "Excess late flights" is an estimate that assumes the peer level is achievable.
8. The relationship between ground time and recovery is strong and consistent across airlines, but it is not proof of causation.

Appendix A — Validation tests

#	Test	Result
1	Ingestion checks per month (period, duplicates, cancellation codes, client airlines present)	12 of 12 months OK
2	Delay-cause minutes reconcile to arrival delay (Directive #40, sec. III.5)	177,012 of 177,012 late flights in July 2025 (100%)
3	Row counts reconcile across analyses	631,428 July rows = ingestion log; airline totals = test 2
4	Construct load reconciles to Analyze	4,471,010 client flights in both phases
5	Excess recomputes from components	$158,920 \times 16.446\% = 26,136$ (AA at DFW)
6	Monthly and annual controllable gaps agree	AA at DFW: +16.4 in both
7	Assumption "pre-09:00 departure = first flight"	Rejected (43%–89%); metric changed
8	Dashboard exports complete	19, 60, 5, 30 and 5 rows as designed

Appendix B — Main analysis decisions

Decision	Rule
Rate bases	Late rates use operated flights; cancellation rates use scheduled flights
Missing by design	Empty cause fields on on-time flights and empty cancellation codes on operated flights are left empty
Controllable	Code A + CarrierDelay; LateAircraftDelay reported separately
Peers	Pooled average of the other client airlines departing the same airport
Minimum volume	100+ flights for any result used in conclusions
Aircraft chains	Same Tail_Number and date, ordered by scheduled departure; valid only if the previous arrival airport equals the next departure airport; ground time 1–360 min
Time bands	3-hour bands by scheduled local departure; "before 09" covers 00:00–08:59

The complete list (D1–D20) is in ``docs/cleaning_log.md``.

Appendix C — Full controllable benchmark (12 months)

Airport	Airline	Operated	Ctrl late %	Peers %	Gap	Excess
DFW	AA	158,920	25.2	8.8	+16.4	26,136
DEN	WN	87,968	20.5	12.0	+8.6	7,542
BOS	B6	36,425	17.1	12.3	+4.8	1,736
BOS	WN	7,070	21.0	13.5	+7.5	530
ATL	WN	17,530	15.0	13.1	+1.9	340
ATL	B6	2,487	16.5	13.2	+3.4	84
DEN	B6	1,210	16.7	15.5	+1.2	15
DFW	B6	352	14.8	23.3	–8.5	–30

Airport	Airline	Operated	Ctrl late %	Peers %	Gap	Excess
ATL	AA	6,643	12.5	13.2	-0.7	-44
DEN	AA	8,659	13.8	15.5	-1.7	-147
ATL	DL	236,353	13.2	13.3	-0.1	-322
ATL	UA	7,396	8.9	13.3	-4.4	-326
BOS	AA	16,438	12.3	14.4	-2.1	-341
BOS	UA	12,641	9.0	14.8	-5.9	-740
BOS	DL	25,807	11.5	15.0	-3.4	-890
DEN	DL	11,392	7.5	15.9	-8.4	-953
DFW	UA	9,644	9.1	24.1	-15.0	-1,450
DFW	DL	11,420	8.4	24.3	-15.9	-1,816
DEN	UA	106,528	12.2	18.6	-6.4	-6,790

Gaps and excess are computed from unrounded values.

Appendix D — Repository contents

File	Purpose
01_Plan.md	Business task, scope, metrics, hypotheses and change log
01_download_bts.py	Reproducible ingestion from BTS
02_Analyze.ipynb	Validation, exploration and 12-month all-cause benchmark
03_Construct.ipynb	Controllable benchmark, recovery and time-of-day analysis, exports
results/	Five tables behind the dashboard
docs/ingestion_log.csv	Monthly data-quality log
docs/cleaning_log.md	Validation tests and decisions D1-D20
docs/data_dictionary.md	Columns, codes, metrics and glossary

Sources: U.S. DOT, Bureau of Transportation Statistics — Technical Reporting Directive #40, On-Time Performance (effective January 1, 2026); BTS TranStats, Reporting Carrier On-Time Performance (July 2025 – June 2026).